

Analysis I by Terence Tao

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Chapter 1

Natural numbers

1.1 The Peano axioms

Peano axiom is a way to define the natural numbers, an informal definition would be:

Definition 1.1.1: Peano axioms (informal)

A natural number is any element of the set

$$N := \{0, 1, 2, \dots\},$$

which is the set of all the numbers created by starting with 0 and then counting forward indefinitely. We call N the set of natural numbers.

Note:-

Some text books doesn't include the 0 and start with 1. Natural numbers are sometimes called whole numbers.

Definition 1.1.2: Increment

We define increment operation, on a number n by

$$n ++$$

where, for example $3 ++ = 4$.

Definition 1.1.3

Axioms:

1. 0 is a natural number
2. If n is a natural number, so is $n ++$.

Definition 1.1.4

We define 1 to be $0 ++$, 2 to be $(0 ++) ++$, etc.

Proposition 1.1.1

3 is natural number

Proof: By axiom 1), 0 is a natural number. By axiom 2), $0++=1$ is a natural number. By 2) again, $1++=2$ is a natural number. Repeat, $2++=3$ is a natural number. ☺

However, there could be case where, for example, we define $3++=0$, and it loops back to 0. So, there are only 0, 1, 2, 3 in natural numbers and it also does satisfies axiom 1 and 2. Therefore we need to add axiom 3):

Definition 1.1.5

0 is not the successor of any natural number, i.e. $n++ \neq 0, \forall n \in \mathbb{N}$.

There could also be cases where $4++=4$, which doesn't contradict with axioms 1,2,3. Or $4++=1$, which loops back but doesn't contradicts. Thus we add 4)

Definition 1.1.6

Different natural numbers must have different successors; i.e. if $n, m \in \mathbb{N}, n \neq m$, then $n++ \neq m++$. Equivalently, $n++=m++ \Rightarrow n=m$.

Now, we are still missing the steps to construct $\mathbb{N}$ by incrementing. We now add 5):

Definition 1.1.7: Principle of mathematical induction

Let $P(n)$ by any property pertaining to a natural number n . Suppose $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(n++)$ is also true. Then $P(n)$ is true for every natural number n .